

RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

Academic Year: 2020-2021 (Odd Semester)

Innovative Practices- Virtual Laboratory Session Handling Details

Name of the Faculty	: Mr. S. Valai Ganesh
Subject Code and Title	: ME8099 Robotics
Year and Branch	: IV year Mechanical Engineering
Date	: 26.09.2020
Topics Covered	: Robot Kinematics
Name of the Laboratory	: Mechanism and Robotics Laboratory
Website Link	: http://vlabs.iitkgp.ernet.in/mr/#

The PUMA 560 is an industrial robot arm with six degrees of freedom and all rotational joints. In this experiment at first a brief theory about PUMA 560 robot is presented in the theory section. The theory for mathematical computations was obtained from a wide variety of sources encompassing books, papers and internet. In simulation section a virtual model is developed in JavaScript program which is used to investigate the forward kinematics problem.

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Theory

MOVEMASTER RM-501

The Mitsubishi Movemaster RM-501 robot is a robotic arm having five degrees of freedom with five rotational joints. The robot unit consists of the waist, shoulder, elbow, wrist, **pitch, and wrist** joint. These are actuated with DC motors to each of which two encode[rs] are associated which allow differential directional information to be read for each joint.

Overall size:

The Robot Unit, including the base, weighs approximately 27 kg. The overall size of the robot is comparably small with a circular work envelope of 445mm maximum horizontal reach and a maximum payload of 1.2kg.

Components:

In addition to the robot itself, the complete robotic system for the Mitsubishi Movemaster consists of several components, which are shown in the following Figure 2:

Figure 2: A diagram showing the components of the robotic system. It includes a **ROBOT** arm with a **GRIPPER** at the end, connected to a **DRIVE UNIT**, which is in turn connected to a **KINEMATIC CONTROLLER**.

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Virtual Labs
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3D Robot Simulator

Input Panel

Input Joint Values

θ_1	0
θ_2	-45
θ_3	-45
θ_4	0
θ_5	0

OK

Final Transformation Matrix
(End-effector position and orientation)

0.115	-0.307	0.945	430.004
-0.536	-0.820	-0.201	91.400
0.837	-0.483	-0.259	458.902
0.000	0.000	0.000	1.000

Slider-based Robot Joints Controller Panel

Waist Joint: -12

Shoulder Joint: -30

Elbow Joint: -45

Wrist Bend: 0

Wrist Roll: 30

HOME WORKSPACE CLEAR PLOT

Robot end-effector Position

700
600
500
400
300

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INTRODUCTION THEORY OBJECTIVE PROCEDURE SIMULATION QUIZZES REFERENCES

Quizzes

Test Your Knowledge!!

You have scored 5 out of 5.
Your level is: Jeopardy Ready

1. What is the number of degree of freedom (DOF) in Movemaster RM-501

(a) 5
(b) 7
(c) 6
(d) None of this

You have chosen option (a). The answer is correct.

2. How many joints are there in the wrist of the Movemaster RM-501 robot

(a) 2
(b) 4
(c) 3
(d) 1

You have chosen option (a). The answer is correct.

3. What is the P_x positional value of the Movemaster robot in a configuration [60, -45, -47, 0, 0]

(a) 185.282
(b) 320.917 m
(c) 563.377 m
(d) -120.219 m

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